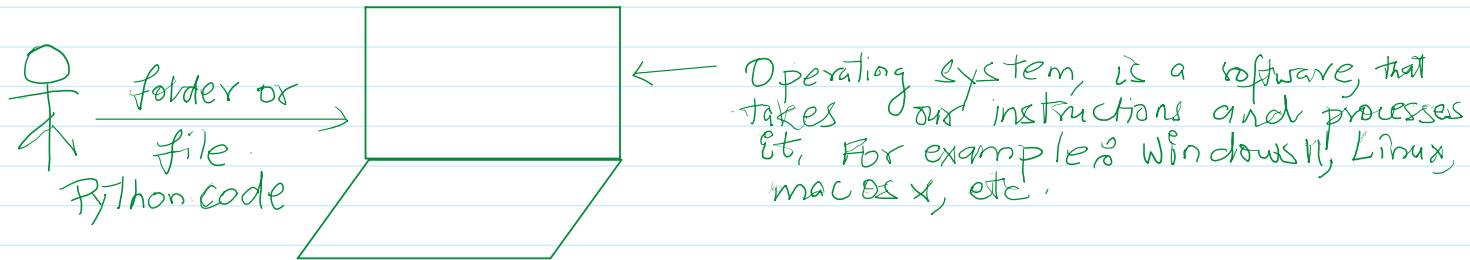


"os" module of Python

06 May 2026 06:49

What is the os module?

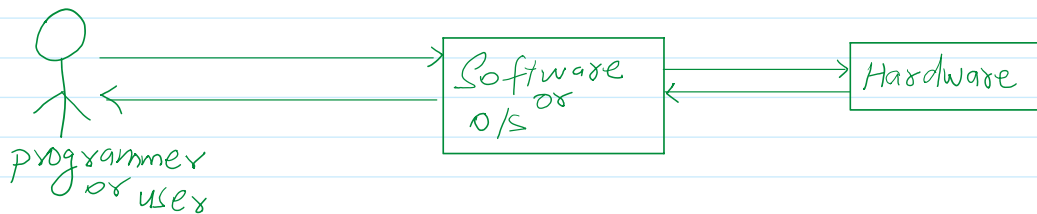
The OS module in Python provides a way to interact with the operating system.



OS help us to create file or folder on it.

Folder is container of inter-related files.

File is a container of data.



"os" module allows your Python program to:

- Work with files and directories.
- Get system information.
- Execute system commands.
- Handle parts. (cross - platform)

Why use the os Module?

Without OS, Python is limited to basic operations. With it, you can:

- Create or delete folders.
- Navigate directories.
- Access environment variables.
- Run shell commands.
- Build platform-independent programs.

Importing the module

```
import os
```

Working with directories(folder)

```
import os  
print(os.getcwd())
```

Output:

Current Working Directory.

```
print(os.getcwd())
```

✓ Current Working Directory.

Output:

```
C:\Users\bluej\Documents\Students\VaanyaSharmaPearsonMiddleSchool\Coding
```

Change Directory

```
import os
print(os.getcwd())
workingDirectory = os.getcwd()+'/learningosmodule'
os.chdir(workingDirectory)
print(os.getcwd())
```

Output:

```
C:\Users\bluej\Documents\Students\VaanyaSharmaPearsonMiddleSchool\Coding
```

```
C:\Users\bluej\Documents\Students\VaanyaSharmaPearsonMiddleSchool\Coding
```

```
\learningosmodule ← changed to learningosmodule.
```

List Files/Folder

```
import os
print(os.getcwd())
parentDirectory = os.getcwd()
workingDirectory = os.getcwd()+'/learningosmodule'
os.chdir(workingDirectory)
print(os.getcwd())
listOfDirOrFiles = os.listdir()
print(listOfDirOrFiles)
print(f'count of files: {len(listOfDirOrFiles)}')
# Moving to the parent directory
# Listing the content of parent directory.
print('Listing the content of parent directory:')
os.chdir(parentDirectory)
listOfDirOrFiles = os.listdir()
# print(listOfDirOrFiles)
for file in listOfDirOrFiles:
    print(file)
print(f'count of files: {len(listOfDirOrFiles)}')
```

Output:

```
['one.py']
count of files: 1
Listing the content of parent directory:
.vscode
Chapter0
Chapter1
Chapter3
Chapter4
Chapter5
Chapter6
```

Chapter7
CSVFileHandling
FileHandling
learningosmodule
StudentManagementSystem
count of files: 12

Create a Directory or Folder

`os.mkdir('newFolder')` ← This create 'newFolder' directory in our current working directory.

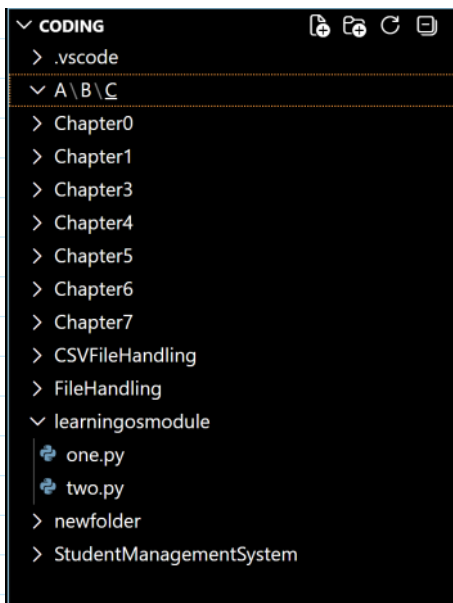
Create Multiple Directories

`import os`
`os.makedirs('A/B/C')` ← It created subdirectories in it.

Output:

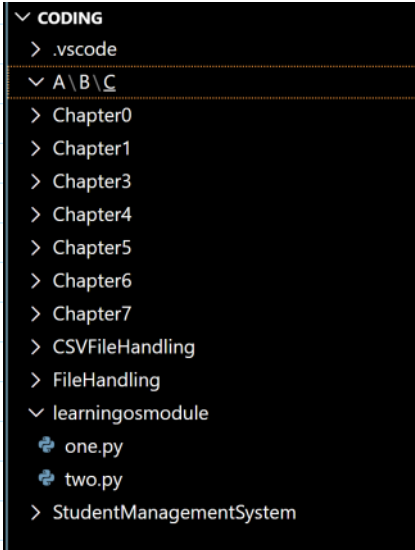


Remove Directory



`import os`
`os.rmdir('newfoder')`

Output:



`os.removedirs('A/B/C')` ← remove all the sub-directories along with parent directory.

Working with files

Rename file

```
os.rename("old_file_name", "new_file_name")
```

Delete File

```
os.remove("file_name")
```

Path Handling(Very Important)

Use `os.path` for safe operations

Join Paths

```
import os
cwd = os.getcwd()+'\\'
path = os.path.join(cwd+'learningosmodule','three.py')
print(path)
with open(path, 'w') as fo: } ← This will create the 'three.py' on the
    pass                      specified path
```

Check File or Directory

```
import os
cwd = os.getcwd()+'\\'
pth = os.path.join(cwd+'learningosmodule','three.py')
print(pth)
isPathExist = os.path.exists(pth) ← we can check the path by using
if isPathExist:                   exists() method of os.path module.
```

```
print(pth)
isPathExist = os.path.exists(pth) ← we can check the path by using
if isPathExist:                      exists() method of os.path module.
    print('yes')
else:
    print('no')
```

Output:
Yes

Check file or Directory

```
import os
pwd = os.getcwd()
isFile = os.path.isfile(pth)
isDir = os.path.isdir(pwd)
print(isFile)
print(isDir)
print(pth)
print(pwd)
```

Output:
True
True
C:\Users\bluej\Documents\Students\VaanyaSharmaPearsonMiddleSchool\Coding
\learningosmodule\three.py ← is a file.
C:\Users\bluej\Documents\Students\VaanyaSharmaPearsonMiddleSchool\Coding

